

Intelligence Visualization for Wave Energy Power Generation

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ABSTRACT

Ocean waves provide a large amount of renewable energy, and Wave energy converter (WEC) can convert wave energy into electric energy. This paper proposes a visualization platform for wave power generation. The platform can monitor various indicators of wave power generation in real time, combined with Long Short-Term Memory (LSTM) neural network to predict wave power and electricity consumption in real time and visualize monitoring data. The platform can intelligently allocate power generation equipment based on the power generation forecast data to achieve precise matching of power generation and power consumption, thereby improving overall power generation efficiency.

Keywords: Wave power generation, Visualization, Artificial intelligence.

Index Terms: Human-centered computing—Visualization—Visualization techniques—Treemaps; Human-centered computing—Visualization—Visualization design and evaluation methods

1 INTRODUCTION

As traditional fossil energy sources are exploited and exhausted and environmental problems become increasingly prominent, and people's demand for energy continues to increase with the development of science and technology, mankind is looking for more renewable energy power generation technologies to supplement and replace existing power generation methods. Ocean energy is currently a hot topic in renewable and sustainable energy supply systems. Compared with other power generation methods, wave energy power generation has the following significant advantages: ocean waves provide the highest energy density among renewable energy sources (density is about 1,000 times that of wind); Wave energy converter (WEC) has a lower negative impact on the environment during use; waves propagate long distances with little energy loss. According to data [1], the power generation rate of WEC is as high as 57%, while the power generation rate of wind and solar power generation devices is 20%-30%.

Although the use of renewable energy for power generation can reduce carbon emissions and improve environmental quality, it has not achieved satisfactory results in terms of the balance of power supply and demand, energy conversion, and equipment failure rate. In order to improve the reliability of wave power generation, this research proposes an artificial intelligence-based wave energy smart power generation visualization platform. The platform monitors various indicators on the wave power generation end and the power distribution demand side in real time through the Internet of Things terminal, and predict the wave power and electricity consumption in real time through artificial neural network, so as to realize the visualization of the forecast data and the monitoring data. The platform can allocate

power generation equipment based on the power generation forecast data, realize the precise matching of power generation and power consumption, and improve the overall power generation efficiency.

2 RELATED WORK

2.1 Wave Power Generation Forecast

The output power of wave energy generating devices is extremely unstable affected by factors such as wind, temperature, light, etc. Referring to existing studies [2], wave height of the waves (H) and wave energy (F) is the key for energy conversion, and the wave energy can be calculated [3] by the formula $F = 0.49 * H^2 * T$. Thus, generating a predetermined position Wave forecast Becomes very important. In this paper, the Long Short-Term Memory (LSTM) model is used to monitor sea level data through ocean buoys, combined with weather data to make a preliminary prediction of ocean wave height, and then use the obtained wave height data as a training set to predict the power generation.

2.2 Electricity Demand Forecast

Affected by seasonal changes, major festivals and other factors, electricity demand has also shown an uncertain upward and downward trend. When the power generation is insufficient, it will bring a lot of inconvenience to people's production and life, and the feature of inability to store electric energy causes the unused electricity to be directly wasted. Therefore, in addition to wave energy power generation forecasts, this research has carried out electricity demand forecasts in order to achieve a dynamic balance between power generation and electricity consumption.

3 SOLUTION

3.1 Visualization Platform

The visualization platform can monitor the status of wave power generation in real time, and detect abnormal conditions in the process of power generation and transmission in time, so that the staff can obtain visual and intuitive information about the power generation equipment they manage, thereby greatly improving the labor efficiency of the staff, management efficiency and capabilities. According to the power generation forecasting and electricity demand forecasting, the platform can remotely control the operation of power generation equipment, thereby improving the overall operating efficiency of the power generation system and achieving a balance between supply and demand.

3.1.1 Energy Monitoring

The functions of each module are shown in the Table 1. Through these modules, the wave energy state can be monitored clearly and intuitively in real time, and discovers the abnormal state of wave energy production in time. Figure 2 shows the visualization of energy monitoring.

3.1.2 Equipment Monitoring

The functions of each module are shown in Table 2. Through these modules, the equipment status can be monitored clearly and intuitively in real time, and the abnormal status of the equipment can be found in time. Figure 3 shows the visual effect of equipment monitoring.

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Figure 1: Energy Information Monitoring Interface

Table 1: The Function Module in Energy Monitoring

Module	Features
System info	Send the system information in real time
Power generation	Display the total power generation of all generators and the utilization efficiency of each generator
Wave forecast	Predict the wave height and compare it with the actual wave height monitored
Equipment efficiency	Real-time monitoring of equipment efficiency
Energy demand	Show the energy demand of the grid at various times of the day
Load condition	Real-time monitoring of the load of the system to avoid overload
Energy conversion rate	Display the current energy conversion rate of the system
Energy distribution	Shows the energy distribution of each device

3.2 VR Remote Control

Another goal of the visualization platform is to create digital twins of a wave power plant in a computer in the physical world, allowing users to remotely view the factory through VR glasses and operating handles. Our equipment is HTC Vive, which is a popular virtual reality head-mounted display product. We built a wave energy power plant model based on Unity3D. Users can view the operating status of the physical hardware in the virtual scene, and control the virtual body in the virtual environment by operating the handle.

At the same time, we used Unity3D to model the working status of wave power equipment at sea, and users can intuitively view the movement of the power equipment through the model. We also want to acquire the real motion of the wave equipment, but this requires the data support of the sensors on the wave energy power generation equipment. The corresponding sensor development for this data has not yet been carried out, so we can only imitate the power generation process. In our future work, we will consider collecting real data feedback on the equipment.

4 CONCLUSION

With the development of economy and society, people are looking for more renewable energy power generation technologies to supplement and replace existing power generation methods. Related

Table 2: The Function Module in Energy Monitoring

Module	Features
Equipment	3D models show the operation of each device in the system
Equipment status	Real-time monitoring of the operating status of each power generation equipment
Energy consumption	Display the current energy consumption
Running record	Record system operation records for the past seven days
Network transmission	Monitor the network status of the equipment and discover network abnormalities in time



Figure 2: View Wave Power Generation Center in Real Time

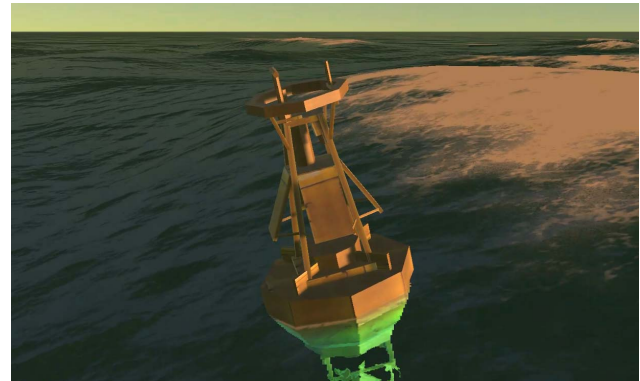


Figure 3: Power Generation Process Model

researches have shown that wave energy power generation has obvious advantages over other renewable energy power generation. Therefore, this research proposes a visualization platform for wave power generation, and the platform's LSTM forecast model can predict of wave power generation and electricity demand with high accuracy. With the help of a visual interface, the platform can monitor wave power generation status and power transmission process in real time. Through the visualization of the whole process, the platform can improve the efficiency of power generation operation and the power generation process stability.

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